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(54) HAIR FASTENERS

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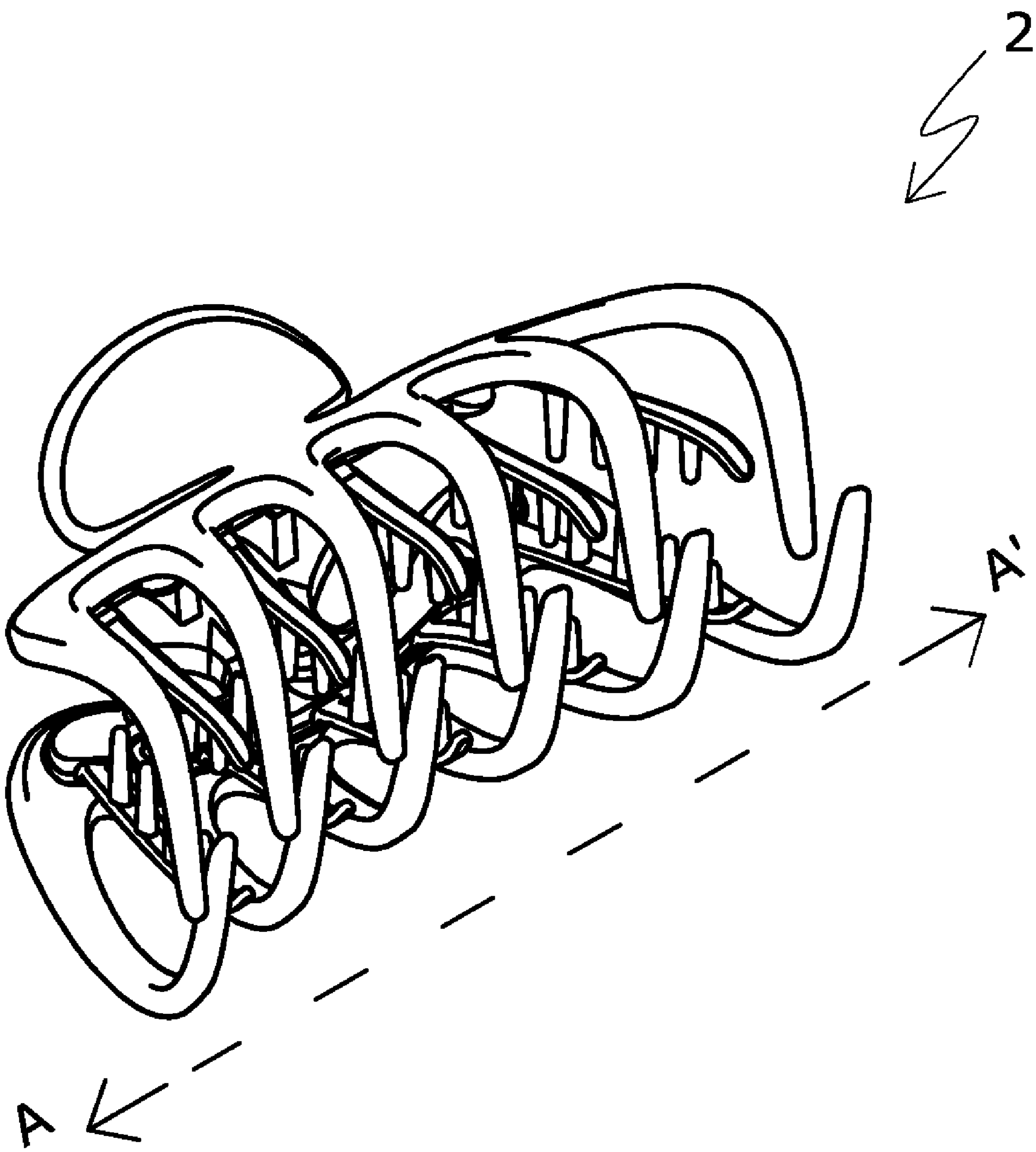
(57) **ABSTRACT**

There is provided a hair fastener in the form of a hair claw device. The hair fastener has a first outer claw member and a second outer claw member. The first and second outer claw members are pivotably connected at and movable about a rear end thereof, and are provided with a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members. The hair fastener further has a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively.



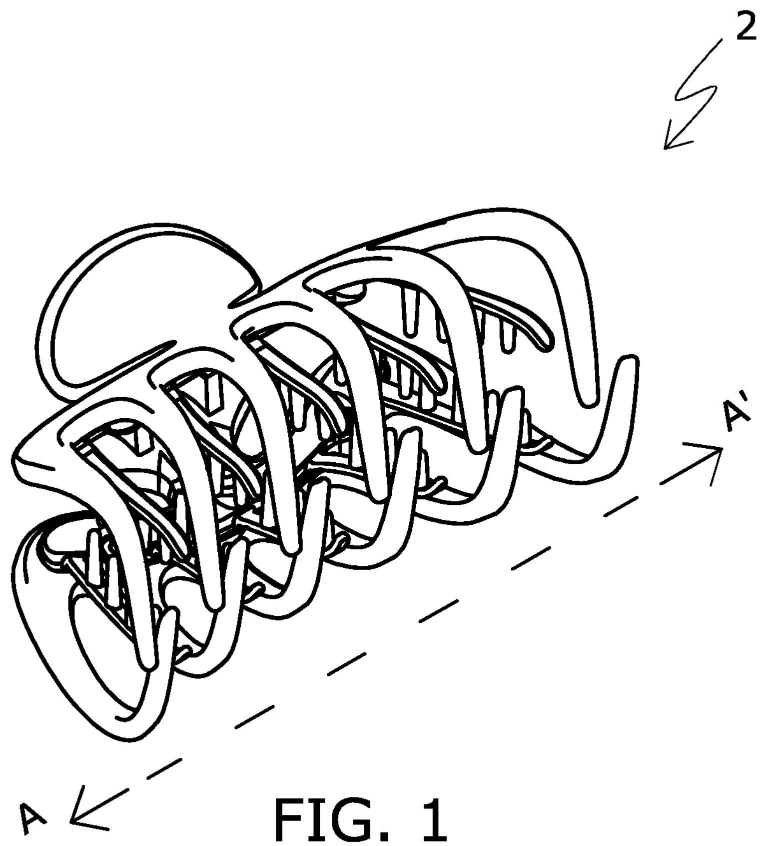


FIG. 1

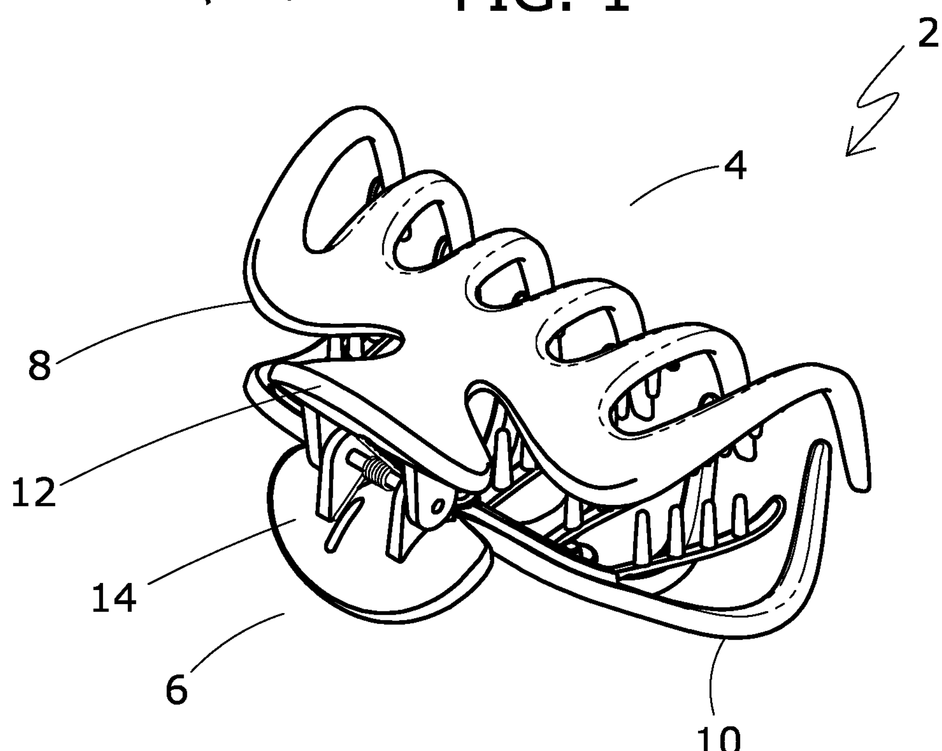
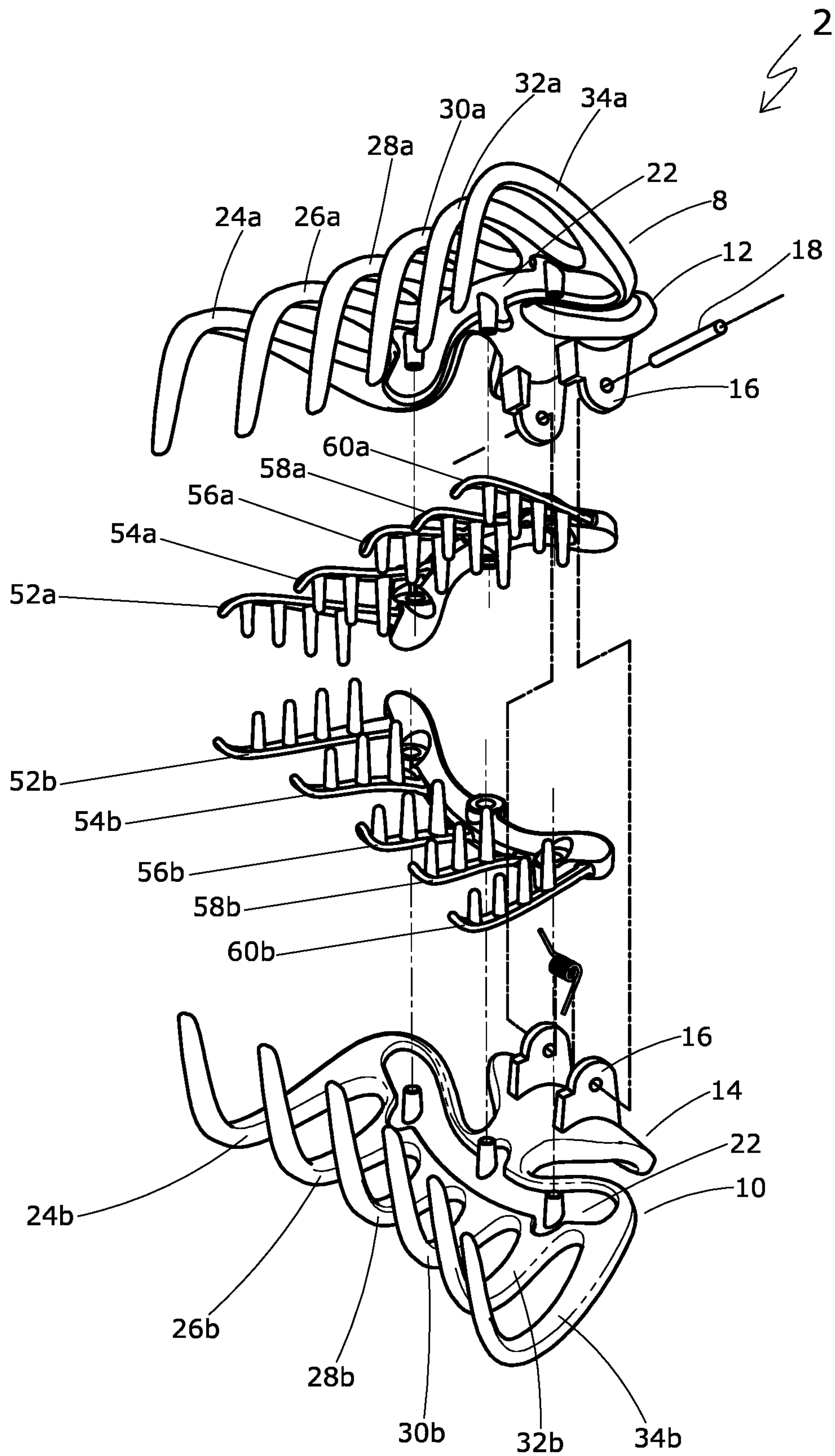


FIG. 2


2
32a
34a
30a
28a
22
24a
26a
8
12
18
16
60a
58a
56a
54a
52a
52b
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32b
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FIG. 3

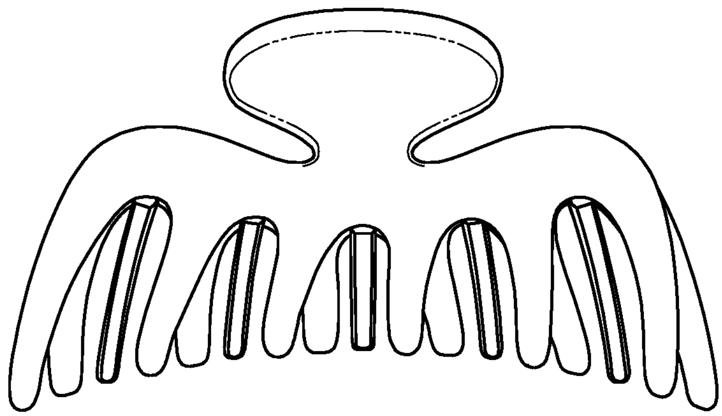

FIG. 4

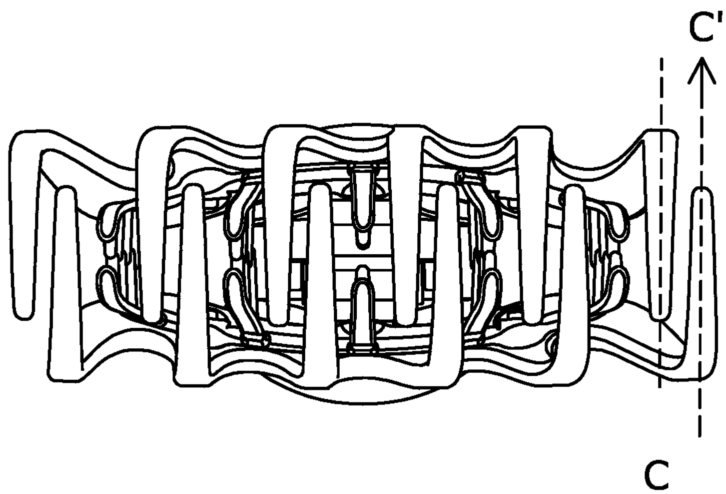

FIG. 5

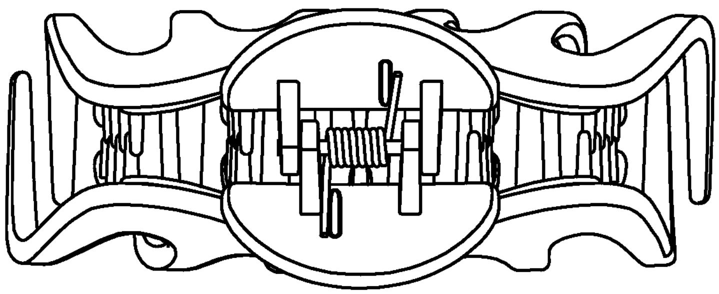

FIG. 6

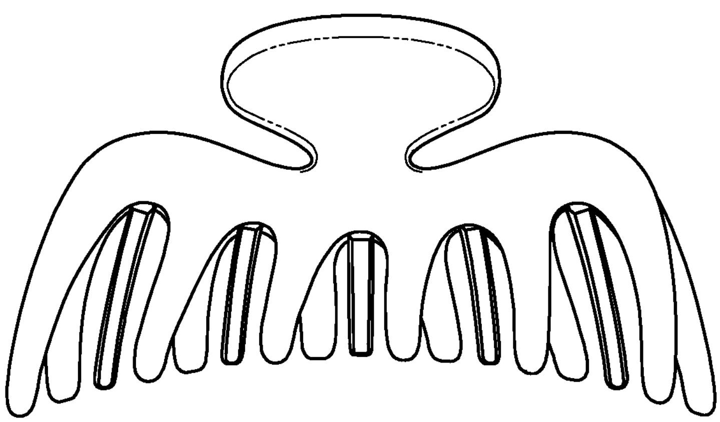

FIG. 7

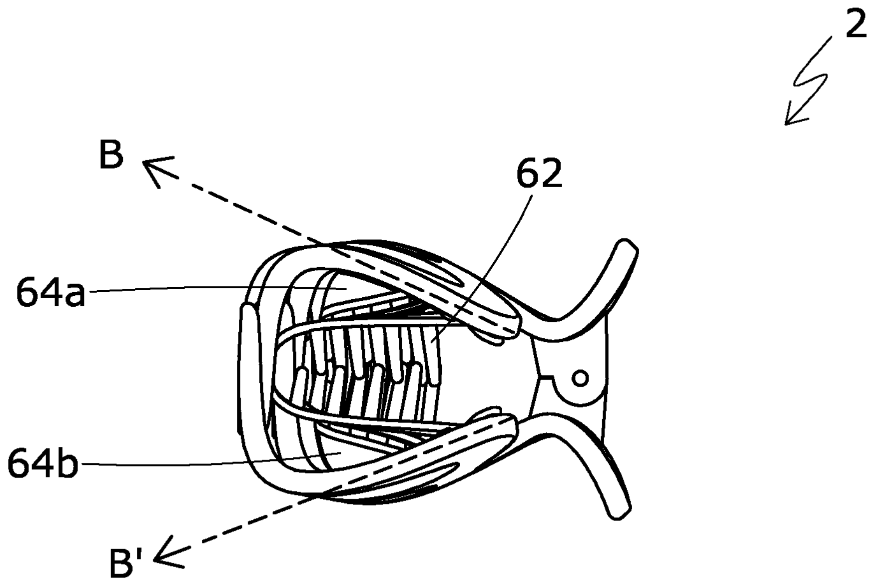

FIG. 8

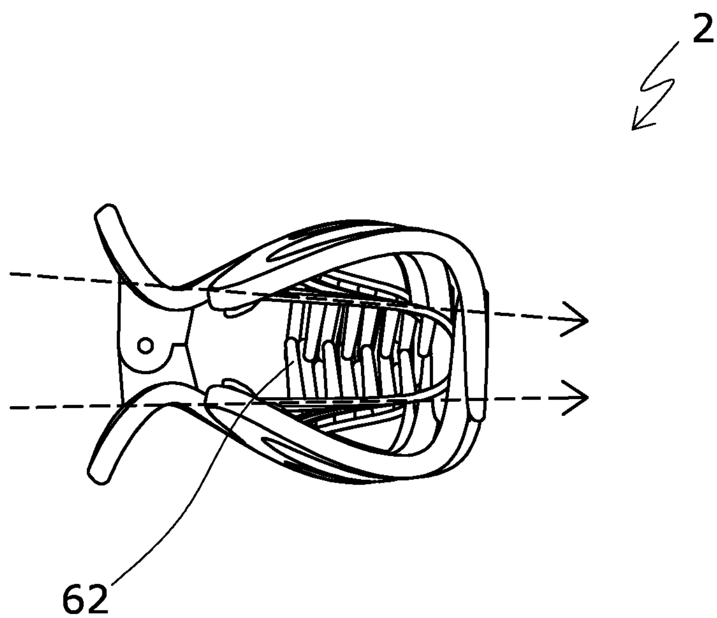

FIG. 9

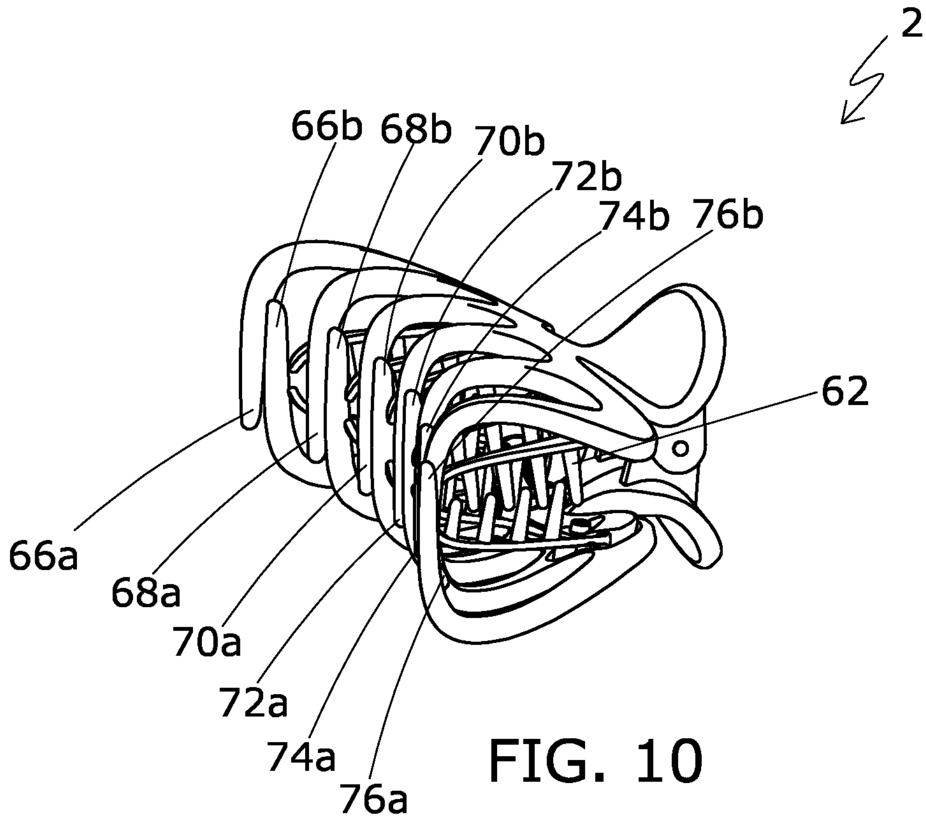

FIG. 10

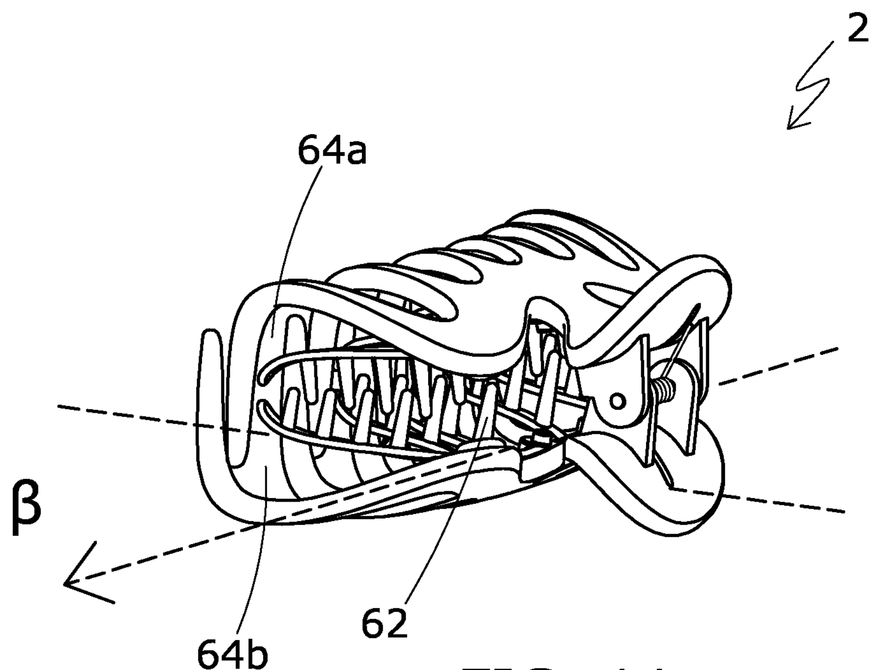

FIG. 11

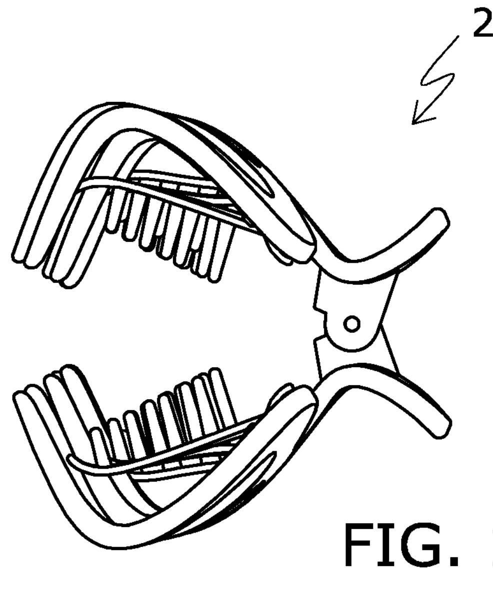

FIG. 12

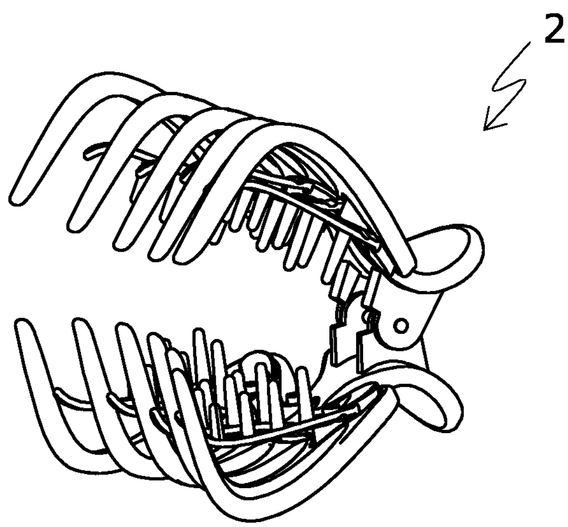

FIG. 13

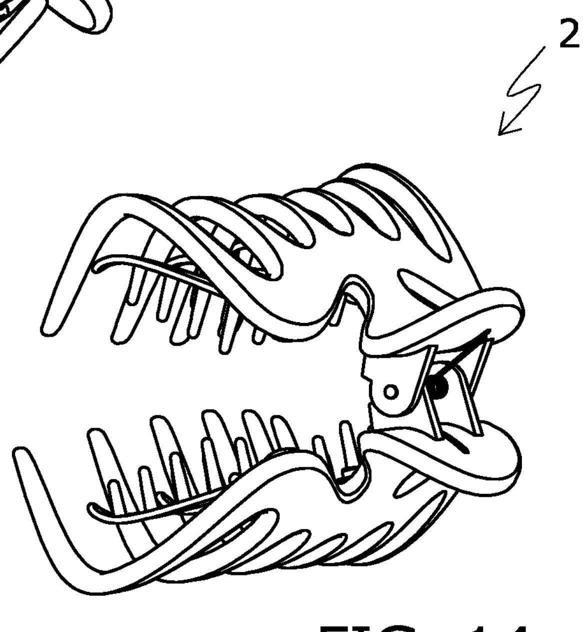

FIG. 14

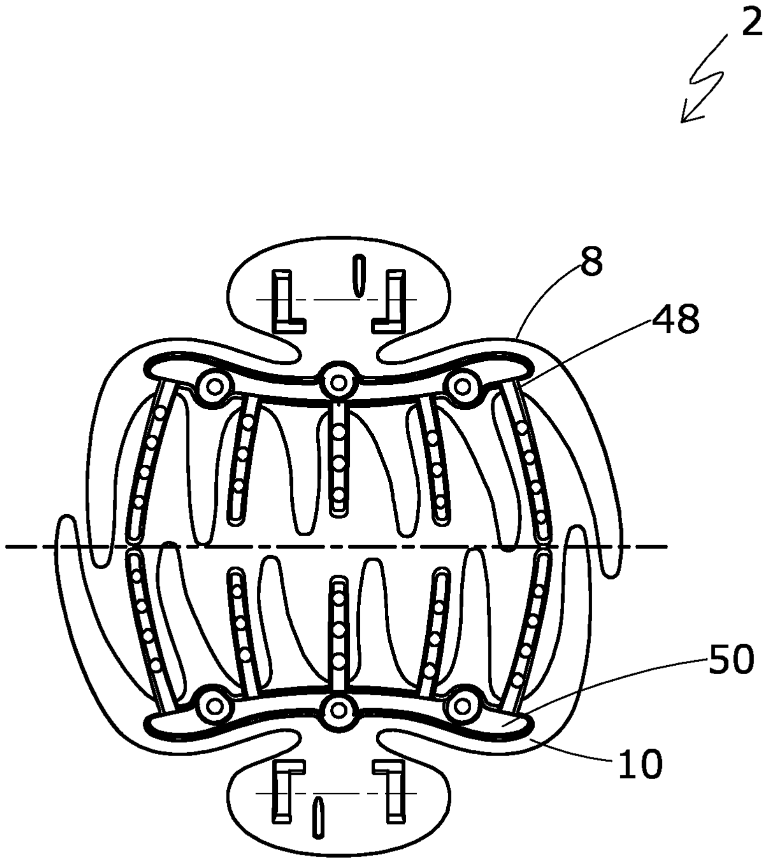

FIG. 15

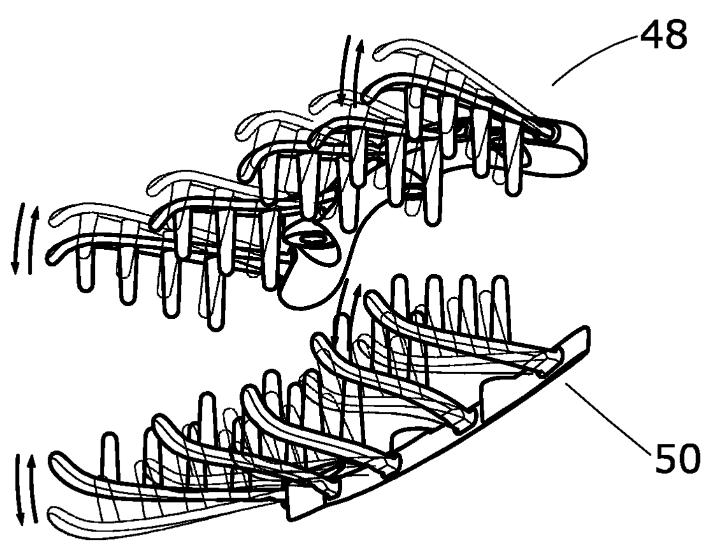

FIG. 16

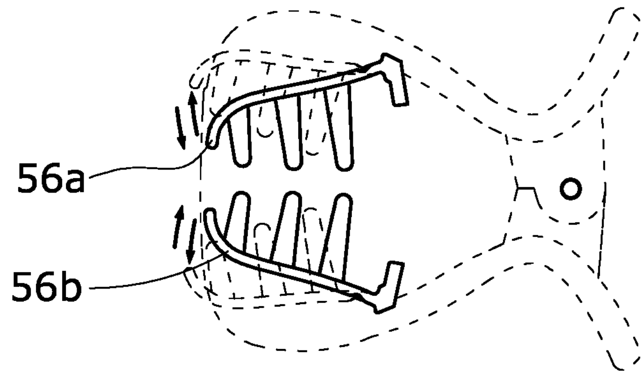

FIG. 17

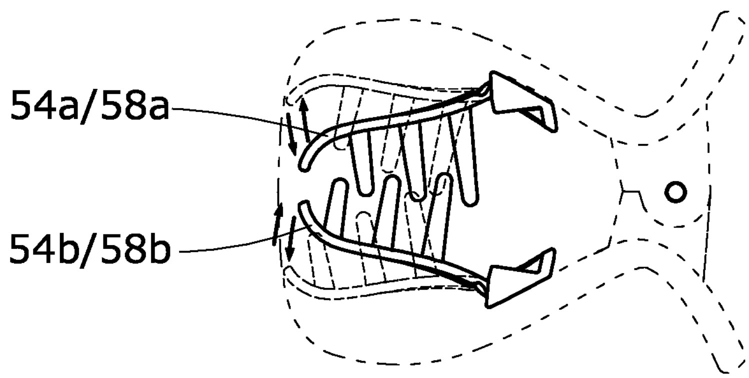

FIG. 18

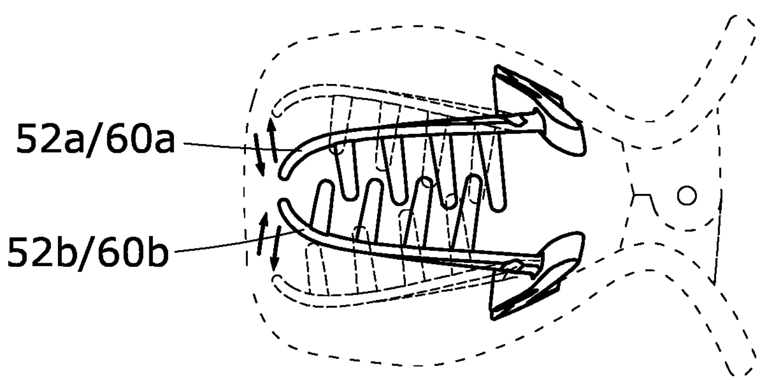

FIG. 19

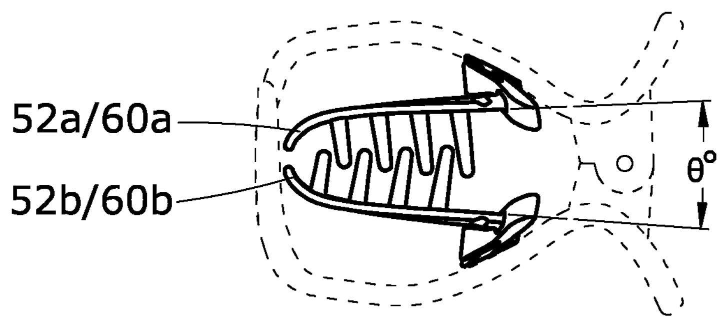

FIG. 20

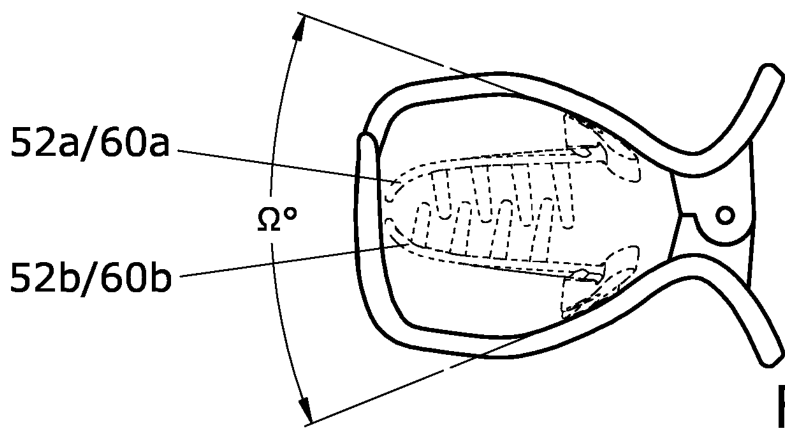

FIG. 21

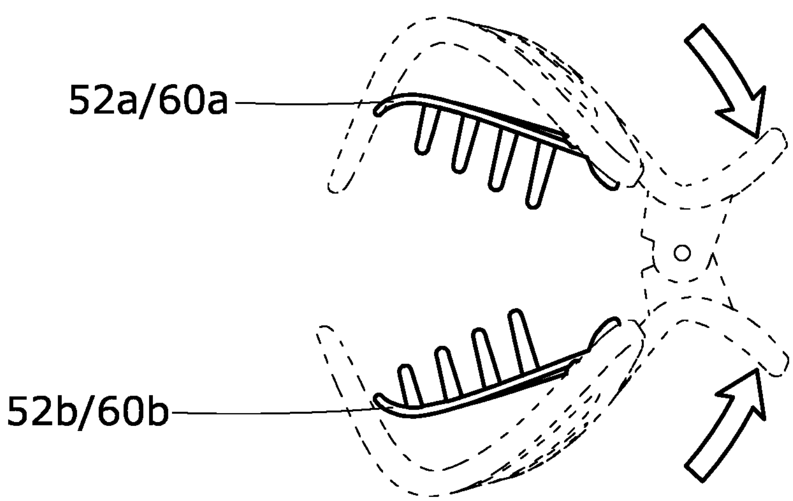

FIG. 22

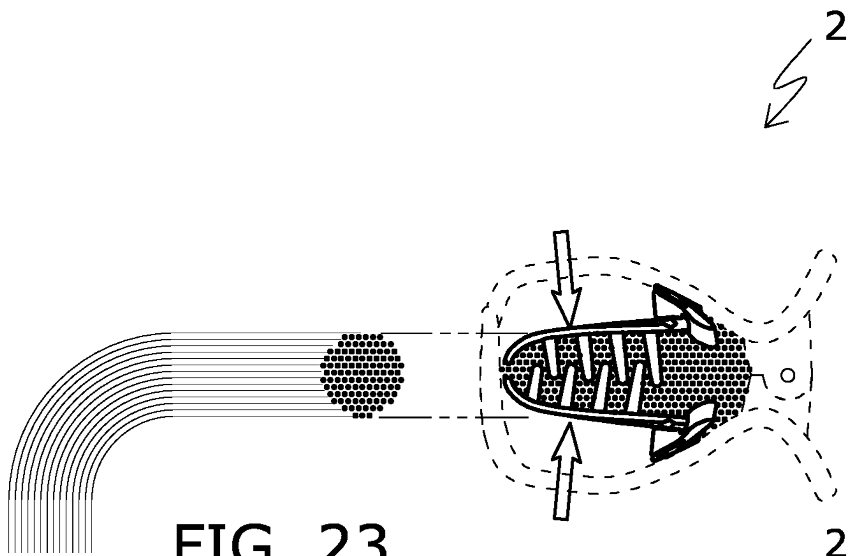

FIG. 23

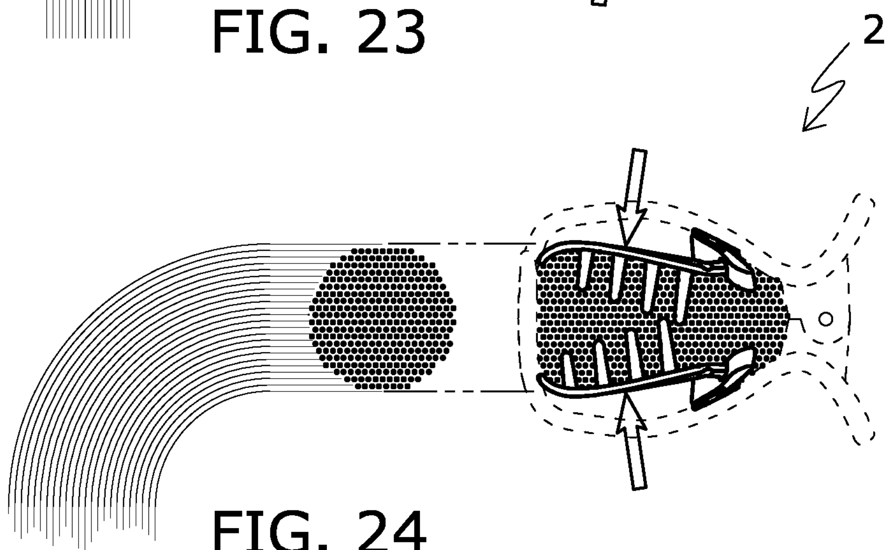

FIG. 24

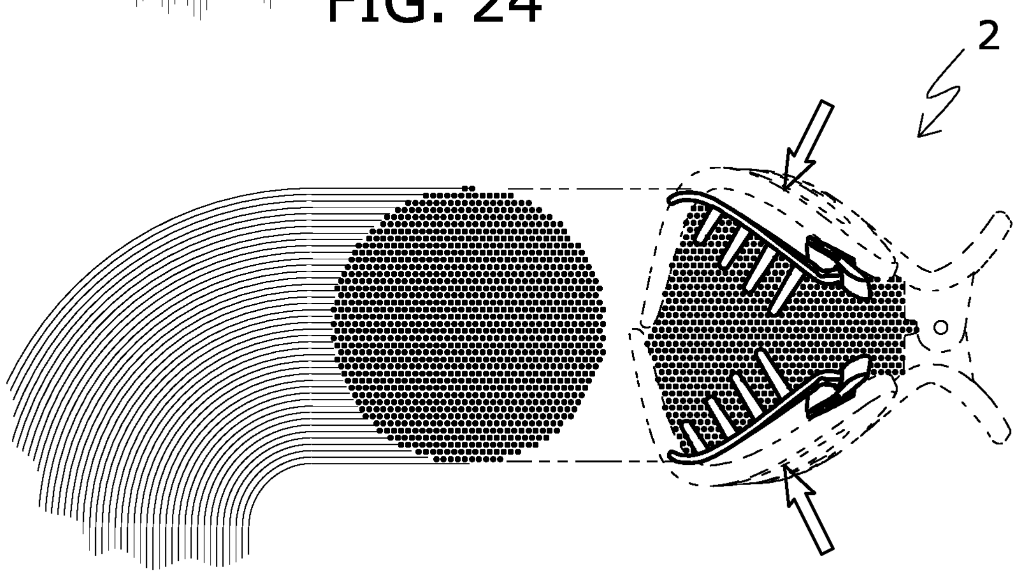

FIG. 25

FIG. 26

HAIR FASTENERS

TECHNICAL FIELD OF THE INVENTION

[0001] The present invention is concerned with hair fasteners and more specifically concerned with hair fasteners often addressed as hair claws.

BACKGROUND OF THE INVENTION

[0002] There is a variety of hair fasteners. Some hair fasteners such as hair barrettes are designed to secure a thinner layer of hair, while some hair fasteners such as a conventional hair loop are often used to secure a thicker lock of hair such as a pony tail. Regardless of the type of hair fasteners concerned, users often find different types of hair fasteners suffer from inadequate strength in securing hair. Users also find that in one styling exercise they would need to use different types of fasteners to secure different thickness of hair.

[0003] The present invention seeks to address the aforementioned problems of conventional hair fasteners, or at least to provide an alternative to the public.

SUMMARY OF THE INVENTION

[0004] According to a first aspect of the present invention, there is provided a hair fastener comprising a first outer claw member and a second outer claw member, wherein the first and second outer claw members are pivotably connected at and movable about a rear end thereof, a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members, the plurality of outer fingers transitions to respective plurality of prongs such that in a default configuration the prongs of the first claw member and the prongs of the second claw member point at opposite directions and assume an interlocking or alternating formation, and the hair fastener further comprises a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively.

[0005] Preferably, the first and second outer claw members may be provided with a pair of handles at the rear end for operating the hair fastener, the first inner claw member and the second inner claw member may be provided with a plurality of inner fingers, and in the default configuration, convergingly and forwardly extend to the front portion of the first and second outer claw members, and at least some of the inner fingers from the first inner claw member may be provided with protrusions and at least some of the inner fingers from the second inner claw members are provided with corresponding protrusions and the protrusions of the first inner claw members and the respective protrusions of the second claw member generally point towards each other and assume a cooperating formation. Cooperating formation refers to the protrusions are sized and shaped and work together to secure a target lock/layer of hair. In preferred embodiments, in the default configuration, the protrusions from the first inner claw member and the protrusions from the second inner claw members may arranged in an alternating manner.

[0006] Suitably, in the default configuration, the plurality of prongs defines a front outwardly facing surface of the front portion of the hair fastener, the outer fingers and the prongs together define a cavity in or between which the first and second inner claw members reside, and/or a distal end of the inner fingers terminates at the front outwardly facing surface of the front portion of the hair fastener.

[0007] In an embodiment, the number of the outer fingers from the first outer claw member may be n, and the number of the inner fingers from the first inner claw members may be n−1. For example, n may be 6.

[0008] Advantageously, the outer fingers from the first outer claw member and the respective inner fingers from the first inner claw member may be arranged laterally across the hair fastener in an alternating manner, and the outer fingers from the second outer claw member and the respective inner fingers from the second inner claw member may be arranged laterally across the hair fastener in an alternating manner. At the default configuration, the inner fingers and/or the protrusions of the first inner claw member may abut at least some of the inner fingers and/or the protrusions of the second inner claw member.

[0009] Preferably, a clearance may be provided between the inner fingers of the first inner claw member and the outer fingers of the first outer claw member, and/or a clearance may be provided between the inner fingers of the second inner claw member and the outer fingers of the second outer claw member, such that the inner fingers of the first inner claw member may flex towards or away from the outer fingers of the first outer claw member and/or the inner fingers of the second inner claw member may flex towards or away from the outer fingers of the second outer claw member.

[0010] In preferred embodiments, by way of dimensional and/or material characteristics, the inner fingers may be more flexible than the inner fingers, the flexural strengths of the outer fingers and the inner fingers are 40-120 MPa and 43.5-130.5 MPa, respectively, and the pull forces of the outer fingers and the inner fingers are 1-16 lbs and 0.2-1.5 lbs, respectively. The first and second outer claw members and the first inner claw members may be made of different materials, and/or the first and second outer claw members are made of a material selected from the group consisting of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), polystyrene (PS), polycarbonate (PC) and high impact polystyrene (HIPS), and the first and second inner claw members are made of a material selected from the group consisting of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers/(SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), and polycarbonate (PC). The more flexible inner fingers may provide more flexibility in accommodating hair locks of different thickness or size to be secured, while the stiffer outer fingers may provide an better insulation to surrounding disturbance to the inner fingers and the hair secured therein.

[0011] Preferably, the angle of inclination of a longitudinal axis defined by one of the inner fingers with respect to a longitudinal axis defined by a respective adjacent outer finger may range from 10° to 45°.

[0012] According to a second aspect of the present invention, there is provided a hair fastener comprising a first outer claw member and a second outer claw member, wherein the first and second outer claw members are pivotably connected at and movable about a rear end thereof, a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members, the plurality of outer fingers transitions to respective plurality of prongs such that at a default configuration the prongs of the first claw member and the prongs of the second claw member point at opposite directions and assume an interlocking or alternating formation, the hair fastener further comprises a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively, and a clearance is provided between the inner fingers of the first inner claw member and the outer fingers of the first outer claw member, and a clearance is provided between the inner fingers of the second inner claw member and the outer fingers of the second outer claw member, such that the inner fingers of the first inner claw member can flex towards or away from the outer fingers of the first outer claw member and the inner fingers of the second inner claw member can flex towards or away from the outer fingers of the second outer claw member.

[0013] Preferably, the first inner claw member and the second inner claw member may be provided with a plurality of inner fingers, and at the default configuration, convergingly and forwardly extend to the front portion of the first and second outer claw members, the angle of inclination of one of the inner fingers with respect to a respective adjacent outer finger range from 10° to 45°.

[0014] Suitably, the first and second outer claw members may be provided with a pair of handles at the rear end for operating the hair fasteners, the plurality of prongs may define a front outwardly facing surface of the front portion of the hair fastener, the outer fingers and the prongs together define a cavity in or between which the first and second inner claw members reside, and a distal end of the inner fingers terminates at the front outwardly facing surface of the front portion of the hair fastener.

[0015] In an embodiment, the number of the outer fingers from the first outer claw member may be n, and the number of the inner fingers from the first inner claw members may be n−1. For example, n may be 6.

[0016] Advantageously, the outer fingers from the first outer claw member and the respective inner fingers from the first inner claw member may be arranged laterally across the hair fastener in an alternating manner, and the outer fingers from the second outer claw member and the respective inner fingers from the second inner claw member may be arranged laterally across the hair fastener in an alternating manner. At the default configuration, the inner fingers and/or the protrusions of the first inner claw member may abut at least some of the inner fingers and/or the protrusions of the second inner claw member.

[0017] In one embodiment, by way of dimensional and/or material characteristics, the inner fingers may be more flexible than the inner fingers, the flexural strengths of the outer fingers and the inner fingers are 40-120 MPa and 43.5-130.5 MPa, respectively, the pull forces of the outer fingers and the inner fingers are 1-16 lb and 0.2-1.5, respectively, and the first and second outer claw members may be made of a material selected from the group of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), polystyrene (PS), polycarbonate (PC) and high impact polystyrene (HIPS), and/or the first and second inner claw members may be made of a material selected from the group of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), and polycarbonate (PC).

[0018] According to a third embodiment of the present invention, there is provided a hair fastener comprising a first outer claw member and a second outer claw member, wherein the first and second outer claw members are pivotably connected at and movable about a rear end thereof, a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members, wherein the plurality of outer fingers transitions to respective plurality of prongs such that in a default configuration the prongs of the first claw member and the prongs of the second claw member point at opposite directions and assume an interlocking or alternating formation, the hair fastener further comprises a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively, at least some of the inner fingers from the first inner claw member are provided with protrusions and at least some of the inner fingers from the second inner claw members are provided with corresponding protrusions and generally point towards each other or at opposite directions and assume a cooperating formation, and a clearance is provided between the inner fingers of the first inner claw member and the outer fingers of the first outer claw member, and a clearance is provided between the inner fingers of the second inner claw member and the outer fingers of the second outer claw member, such that the inner fingers of the first inner claw member can flex towards or away from the outer fingers of the first outer claw member and the inner fingers of the second inner claw member can flex towards or away from the outer fingers of the second outer claw member in the respective clearance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Some embodiments of the present invention will now be explained, with reference to the accompanied drawings, in which:

[0020] FIG. **1** is a front perspective view of an embodiment of a hair fastener in a closed or default configuration,

[0021] FIG. **2** is a rear perspective view of the hair fastener of FIG. **1**,

[0022] FIG. **3** is an exploded view of the hair fastener of FIG. **1**,

[0023] FIG. **4** is a top view the hair fastener of FIG. **1**,

[0024] FIG. **5** is a front view the hair fastener of FIG. **1**,

[0025] FIG. **6** is a rear perspective view of the hair fastener of FIG. **1**,

[0026] FIG. **7** is a bottom view the hair fastener of FIG. **1**,

[0027] FIG. **8** and FIG. **9** are opposite side views of the hair fastener of FIG. **1**,

[0028] FIG. **10** and FIG. **11** are alternative perspective views of the hair fastener of FIG. **1**,

[0029] FIG. **12** is an alternative side view of the hair fastener of FIG. **8** but in an open configuration,

[0030] FIG. **13** and FIG. **14** are alternative perspective views of the hair fastener of FIG. **12**,

[0031] FIG. **15** is a schematic view illustrating the arrangement of fingers and prongs of first and second outer claw members and fingers of first and second inner claw members of the hair fastener of FIG. **1**,

[0032] FIG. **16** is a schematic view illustrating first and second inner claw members of the hair fastener of FIG. **1**,

[0033] FIG. **17** is a schematic view showing respective (middle) fingers of the inner claw members of the hair fastener of FIG. **16**,

[0034] FIG. **18** is a schematic view showing respective second and fourth fingers of the inner claw members of the hair fastener of FIG. **16**,

[0035] FIG. **19** is a schematic view showing respective first and fifth fingers of the inner claw members of the hair fastener of FIG. **16**,

[0036] FIG. **20** is a schematic view showing the respective first fingers of FIG. **19**,

[0037] FIG. **21** is a schematic view showing, in the default configuration, the profile and relative position of two side fingers extending from first and second outer claw members of the hair fastener of FIG. **1**,

[0038] FIG. **22** is a schematic view showing the profile and relative position of the two side fingers extending from the first and second inner claw members of the hair fastener of FIG. **21** but in an open configuration,

[0039] FIG. **23**, FIG. **24** and FIG. **25** are schematic views illustrating the use of the hair fastener of FIG. **1** for securing three respective locks of hair with different thicknesses in different scenarios, and

[0040] FIG. **26** is a schematic diagram illustrating the use of the hair fastener of FIG. **1** for securing a layer of hair of a user.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION

[0041] The present invention is concerned with a hair fastener, and is illustrated and explained by way of examples below.

[0042] FIGS. **1-26** illustrate an embodiment of a hair fastener according to the present invention. The hair fastener is generally designated **2**. The hair fastener **2** is in the form of a hair claw device.

[0043] FIGS. **1** and **2** are front and rear perspective views of the hair fastener **2** in a default (closed) configuration. FIGS. **1** and **2** illustrate that the hair fastener **2** generally has an elongate profile and extends laterally. Please see arrows A-A'. The hair fastener **2** has a front end **4** and a rear end **6**. The hair fastener **2** comprises a pair of outer claw members, namely a first (or upper) outer claw member **8** and a second (or lower) outer claw member **10** arranged at the front end **4**. The hair fastener **2** has a pair of cooperating handles **12**, **14** extended from a rear portion of the outer claw members **8**, **10**. In this embodiment, the first and second outer claw members **8**, **10** are pivotably connected at the handles **12**, **14**. In use, the opening or closing of the outer claw members **8**, **10** is operated by the handles **12**, **14** at the rear end **6**. Please also see FIG. **6**. On pressing or bringing the handles **12**, **14** together, the first and second outer claw members **8**, **10** are spread apart.

[0044] FIG. **3** is an exploded view of the hair fastener **2**. In this view, there is illustrated a hinge system. In this embodiment, the hinge system includes two pairs of cooperating lugs **16** provided on and extended from inwardly and oppositely facing surfaces of the handles **12**, **14** and a laterally extending pin **18** that runs through openings on the lugs **16**. The hinge system also includes a torsional spring **20** for biasing the hair fastener **2** or the outer claw members **8**, **10** to the default configuration as shown in, for example, FIGS. **1** and **2**. On release of the handles **12**, **14**, the first and second outer claw members **8**, **10** are biased together. In use, the outer claw members are pivotably movable about the longitudinal axis of the pin **18**.

[0045] Each of the first and second outer claw members **8**, **10** has a base portion **22** from which a plurality of outer fingers (**24***a*, **24***b*, **26***a*, **26***b*, **28***a*, **28***b*, **30***a*, **30***b*, **32***a*, **32***b*, **34***a*, **34***b*) protrudes forwardly and outwardly. The base portion **22** is elongate in shape and laterally extending. Please also see FIGS. **4-5**, **8-9**. In the default configuration, the outer fingers (**24***a*, **26***a*, **28***a*, **30***a*, **32***a*, **34***a*) from the first outer claw member **8** and the outer fingers (**24***b*, **26***b*, **28***b*, **30***b*, **32***b*, **34***b*) from the second outer claw members **10** extend forwardly in a diverging manner. Please also see arrows B, B' in FIG. **8**. Please also see FIG. **21**. The outer fingers (**24***a*, **24***b*, **26***a*, **26***b*, **28***a*, **28***b*, **30***a*, **30***b*, **32***a*, **32***b*, **34***a*, **34***b*) of the first and second outer claw members **8**, **10** at their distal ends however transition to respective plurality of prongs (**36***a*, **36***b*, **38***a*, **38***b*, **40***a*, **40***b*, **42***a*, **42***b*, **44***a*, **44***b*, **46***a*, **46***b*). As shown in for example FIGS. **5**, **7**, **8-9** and **10-11**, in the default configuration, the prongs (**36***a*, **38***a*, **40***a*, **42***a*, **44***a*, **46***a*) from the first outer claw member **8** and the prongs (**36***b*, **38***b*, **40***b*, **42***b*, **44***b*, **46***b*) from the second outer claw member **10** generally point at opposite directions and assumes an interlocking or arranged in an alternating formation. Please see arrows C, C' in FIG. **5** illustrating the orientation and arrangement of the prongs.

[0046] The hair fastener **2** further comprises a pair of a first inner claw member **48** and a second inner claw member **50**. Each of the first and second inner claw members **48**, **50** is provided with a base portion **52** from which a plurality of respective inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) extends forwardly. The base portion **52** is elongate in shape and laterally extending. However, the base portion **52** of the first and second inner claw members **48**, **50** is shorter than the base portion **22** of the first and second outer claw members **8**, **10**. Due to the shorter base portion **22** of the inner claw members **48**, **50**, the base portion of the inner claw members are concealed within the outer claw members **8**, **10**. The inner claw members **48**, **50** are fixedly connected to inwardly and oppositely facing surfaces of the respective outer claw members **8**, **10** at a rear portion thereof. In this embodiment, the base portion **22** is provided with an elongate recess extending laterally across the inner claw member **8/10**, in which the base portion **52** can sit. Further, vertical posts are provided at the elongate recess and respective openings are provided in the base portion **52** such that the base portion **52** can fit securely in the recess with the posts engaged in the respective openings in order to ensure a secure connection. The connection can be enhanced by an interference fit, an adhesive, snap fitting, or stamping distal (upper) ends of he posts to form an enlarged flange to hold the base portion **52** in place at the openings.

[0047] The inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) of the inner claw members **48**, **50** extend forwardly to a distal (free) end. Please see FIG. **3**. In the default configuration, the inner fingers (**52***a*, **54***a*, **56***a*, **58***a*, **60***a*) from the first inner claw member **48** and the outer fingers (**52***b*, **54***b*, **56***b*, **58***b*, **60***b*) from the second inner claw member **50** extend in a converging manner. Please see arrows D, D' in FIG. **9**. Please also see FIG. **20**. At the free (distal) end, the inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) from the first and second inner claw members **48**, **50** are slightly curved to lie adjacent each other so that the free ends of the inner fingers (**52***a*, **54***a*, **56***a*, **58***a*, **60***a*) of the first inner claw **48** members are in close proximity or in abutment with the respective free ends of the inner fingers (**52***b*, **54***b*, **56***b*, **58***b*, **60***b*) of the second inner claw member **50**.

[0048] The inner fingers **48**, **50** are provided with protrusions extending from inwardly facing surfaces thereof. In this embodiment, the protrusions assume the shape of spikes **62** and extending generally perpendicularly from the respective inner fingers **48**, **50**. At least some of the protrusions from the respective first and second inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) actually abuts each other. The protrusions are designed to enhance gripping onto a wearer's hair during use. In the default configuration, the first and second inner claw members **48**, **50** and thus the respective inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) are biased together. Accordingly, the protrusions from the respective inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) are also biased together. Please see for example FIGS. **8-11**.

[0049] The hair fastener **2** is configured such that a clearance **64***a* is provided between the outer fingers (**24***a*, **26***a*, **28***a*, **30***a*, **32***a*, **34***a*) of the first outer claw member **8** and the respective inner fingers (**52***a*, **54***a*, **56***a*, **58***a*, **60***a*) of the first inner claw member, and likewise, a clearance **64***b* is provided between the outer fingers (**24***b*, **26***b*, **28***b*, **30***b*, **32***b*, **34***b*) of the second claw member **10** and the respective inner fingers (**52***b*, **54***b*, **56***b*, **58***b*, **60***b*) of the second inner claw member **50**. Please see for example, FIGS. **8-9**.

[0050] FIGS. **10-11** show that, in the default configuration, at the front end of the hair fastener **2**, the interlocking prongs (**36***a*, **36***b*, **38***a*, **38***b*, **40***a*, **40***b*, **42***a*, **42***b*, **44***a*, **44***b*, **46***a*, **46***b*) from the first and second outer claw members **8**, **10** define a front outwardly facing surface. The outer fingers and the prongs together define a cavity **78** in or between which the first and second inner claw members **48**, **50** reside. The distal end of the inner fingers (**52***a*, **52***b*, **54***a*, **54***b*, **56***a*, **56***b*, **58***a*, **58***b*, **60***a*, **60***b*) terminates at the front outwardly facing surface.

[0051] As described above, the inner fingers and the respective outer fingers are provided with the clearance **64***a*, **64***b* therebetween. The extent or degree of this clearance **64***a*, **64***b* also plays a role in the functionality of the hair fastener **2**. Studies leading to present invention has shown that the angle of inclination (B) of (a longitudinal axis defined by) one of the inner fingers in relation to (a longitudinal axis defined by) a respective adjacent outer finger ranging from 10° to 45° can preferably provide a desirable room for the inner fingers to flex to or away from the respective outer fingers during use. Please see for example FIG. **11**.

[0052] FIGS. **12-14** are different perspective views showing the hair fastener **2** in an open configuration. From these figures, it can be seen that the protrusions closer to the rear end of hair fastener **2** are longer and the protrusions closer to the front end of the hair fastener are progressively shorter.

[0053] FIG. **15** is a schematic diagram further illustrating the arrangement of the first and second inner claw members **48**, **50** on the first and second outer claw members **8**, **10**, respectively. It is demonstrated that (when viewing from above) the outer fingers (**24***a*, **26***a*, **28***a*, **30***a*, **32***a*, **34***a*) of the first outer claw member **8** and the inner fingers (**52***a*, **54***a*, **56***a*, **58***a*, **60***a*) of the first inner claw member **48** are generally laterally arranged in an alternately manner. Likewise, the outer fingers (**24***b*, **26***b*, **28***b*, **30***b*, **32***b*, **34***b*) of the second outer claw member **10** and the inner fingers (**52***b*, **54***b*, **56***b*, **58***b*, **60***b*) of the second inner claw member **50** are laterally arranged in an alternately manner.

[0054] FIG. **15** also illustrates that, in this embodiment, on each of the outer claw members **8**, **10**, there are provided the six (n) outer fingers (**24***a*/**24***b*, **26***a*/**26***b*, **28***a*/**28***b*, **30***a*/**30***b*, **32***a*/**32***b*, **34***a*/**34***b*) with an elongate gap **80** between each pair of adjacent fingers (e.g. **24***a*, **26***b*). The farthest left (first) and farthest right (sixth) outer fingers are longest; the second and fifth outer fingers are shorter; the third and fourth outer fingers are shortest. On each of the inner claw members, there are provided five (n−1) inner fingers. The inner fingers are positioned at or adjacent the respective adjacent finger gaps **80**. The farthest left (first) and farthest right (fifth) inner fingers are longest; the second and fourth inner fingers are shorter; the third finger in the middle is shortest. In other preferred embodiments, there is provided the same number of inner fingers as the outer fingers, or are more inner fingers than outer fingers.

[0055] The hair fastener **2** is constructed such that the outer fingers (**24***a*, **24***b*, **26***a*, **26***b*, **28***a*, **28***b*, **30***a*, **30***b*, **32***a*, **32***b*, **34***a*, **34***b*) and the prongs (**36***a*, **36***b*, **38***a*, **38***b*, **40***a*, **40***b*, **42***a*, **42***b*, **44***a*, **44***b*, **46***a*, **46***b*) are relatively rigid and the inner fingers are more flexible than the inner fingers such that in use the inner fingers can bend more easily in the cavity. In this embodiment, the flexural strengths of the outer fingers and the inner fingers are 80 MPa and 87 MPa, respectively; and the pull forces of the outer fingers and the inner fingers are 1-16 lbs and 0.2-1.5 lbs, respectively. Studies leading the present invention have shown that the workable flexural strengths of the outer fingers and the inner fingers are 40-120 MPa and 43.5-130.5 MPa, respectively, and the pull forces of the outer fingers and the inner fingers are 1-16 lbs and 0.2-1.5 lbs, respectively.

[0056] In this embodiment, both the outer claw members **8**, **10** and the inner claw menbers are made from injection moldable materials although the outer claw members **8**, **10** and the inner claw members **48**, **50** are be made from different materials The studies have also shown that suitable materials of the first and second outer claw members **8**, **10** include acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), polystyrene (PS), polycarbonate (PC) and high impact polystyrene (HIPS), while suitable materials for the first and second inner claw members **48**, **50** include acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers/(SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), and polycarbonate (PC). These materials are suitable partly because they can assist to configure the inner claw members and the outer claw members to possess the desirable flexibility. However, the inner claw members **48**, **50** may also be made from a springy metallic material.

[0057] In preferred embodiments, the width and thickness of the inner fingers are 3.5±2 mm and 1.3±0.3 mm. The lengths of the first and fifth inner fingers, the second and the fourth inner fingers, and the third inner fingers are 30±6 mm, 24±4.8 mm and 20±4 mm, respectively.

[0058] By way of the aforementioned physical (e.g. dimensional) and/or material characteristics, hair fasteners according to the present invention address desired functionalities which will be discussed further below. FIG. **16** illustrates the flexing of the inner fingers with respect to the outer fingers during use.

[0059] FIGS. **17-19** are schematic diagrams showing third (middle) inner **56***a*, **56***b* fingers, second and fourth fingers **54***a*/**58***a*, **54***b*/**58***b*, and first and fifth inner fingers **52***a*/**60***a*, **52***b*/**60***b* of the inner claw members **48**, **50** of the hair fastener **2**, respectively. In this embodiment, the third fingers, second and fourth, and first and fifth inner fingers are different in a number of ways. Firstly, each of the second, third and fourth inner fingers are provided with three protrusions while each of the first and fifth inner fingers are provided with four protrusions. Secondly, in the default configuration, while the protrusions from the opposing third inner fingers point towards each other, the protrusions from the opposing first and second inner fingers point abut each other and form an interlocking pattern.

[0060] FIGS. **2-22** illustrate the profile of the first and fifth inner finger **52***a*/**60***a*, **52***b*/**60***b* located at opposite lateral ends of the hair fastener **2** when in different configurations.

[0061] FIGS. **23-25** illustrate the use of the hair fastener **2** in different hair securing exercises. FIG. **23** shows that when the lock or layer of hair to be secured is relatively thin, the inner claw members can clip onto the hair with minimal or without the opposing inner fingers having to flex towards their respective outer claw members or bend away from each other. It is to noted that due to the relatively thin hair layer or lock, the relatively small cavity defined between the opposing inner claw members can already accommodate the hair layer/lock. FIG. **24** shows that when the lock or layer of hair to be secured is medium in thickness, the opposing inner fingers from the inner claw members can flex outwardly away from each other and towards their respective outer claw members in a small degree or extent in order to increase room for accommodating the medium thickness of hair layer/lock. In this state, there is still some clearance between the inner fingers and the respective outer fingers. FIG. **25** shows that when the lock or layer of hair to be secured is relatively thickest, the opposing inner fingers from the inner claw members flexes all the way such that the clearance is reduced to minimal and the cavity is enlarged to the fullest extent in order to provide most room for accommodating the thickest of hair layer/lock.

[0062] Further, with the construction of a double claw arrangement, i.e. the presence of both outer claw members and inner claw members, the hair fastener **2** can secure hair more tightly for longer period of time without need to reclip. The outer claw members further insulate hair already secured by the inner claw members from loosening or shifting in position.

[0063] FIG. **25** is a schematic diagram showing the use of the hair fastener. While the user has a rich amount of hair, the user chooses to style by securing only a top layer of hair of medium thickness. It is envisaged the user can easily switch the hair style during the day by, for example, grouping her hair into a thicker ponytail and use the hair fastener **2** to secure the pony tail into shape. Alternatively, she can change the hair style by securing a thinner layer at the side. In either case, the hair fastener **2** can adjust and still maintain a good strength in securing the hair accordingly.

[0064] From the above, it is demonstrated that hair fasteners made in according to the present invention is not only structurally different, but also functionally versatile.

[0065] It should be understood that certain features of the invention, which are, for clarity, described in the content of separate embodiments, may be provided in combination in a single embodiment. Conversely, various features of the invention which are, for brevity, described in the content of a single embodiment, may be provided separately or in any appropriate sub-combinations. It is to be noted that certain features of the embodiments are illustrated by way of non-limiting examples. Also, a skilled person in the art will be aware of the prior art which is not explained in the above for brevity reasons.

What is claimed is:

1. A hair fastener comprising:

a first outer claw member and a second outer claw member, wherein the first and second outer claw members are pivotably connected at and movable about a rear end thereof,

a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members,

wherein:

the plurality of outer fingers transitions to respective plurality of prongs such that in a default configuration the prongs of the first claw member and the prongs of the second claw member point at opposite directions and assume an interlocking or alternating formation, and

the hair fastener further comprises a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively.

2. A hair fastener as claimed in claim **1**, wherein:

the first and second outer claw members are provided with a pair of handles at the rear end for operating the hair fastener,

the first inner claw member and the second inner claw member are provided with a plurality of inner fingers, and in the default configuration, convergingly and forwardly extend to the front portion of the first and second outer claw members, and

at least some of the inner fingers from the first inner claw member are provided with protrusions and at least some of the inner fingers from the second inner claw members are provided with corresponding protrusions and the protrusions o the first inner claw members and the respective protrusions of the second claw member generally point towards each other and assume a cooperating formation.

3. A hair fastener as claimed in claim **2**, wherein, in the default configuration:

the plurality of prongs defines a front outwardly facing surface of the front portion of the hair fastener,

the outer fingers and the prongs together define a cavity in or between which the first and second inner claw members reside, and

a distal end of the inner fingers terminates at the front outwardly facing surface of the front portion of the hair fastener.

4. A hair fastener as claimed in claim **2**, wherein the number of the outer fingers from the first outer claw member is n, and the number of the inner fingers from the first inner claw members is n−1.

5. A hair fastener as claimed in claim **4**, wherein n is 6.

6. A hair fastener as claimed in claim **2**, wherein the outer fingers from the first outer claw member and the respective inner fingers from the first inner claw member are arranged laterally across the hair fastener in an alternating manner, and the outer fingers from the second outer claw member and the respective inner fingers from the second inner claw member are arranged laterally across the hair fastener in an alternating manner.

7. A hair fastener as claimed in claim **6**, wherein at the default configuration, the inner fingers and/or the protrusions of the first inner claw member abut at least some of the inner fingers and/or the protrusions of the second inner claw member.

8. A hair fastener as claimed in claim **2**, wherein a clearance is provided between the inner fingers of the first inner claw member and the outer fingers of the first outer claw member, and a clearance is provided between the inner fingers of the second inner claw member and the outer fingers of the second outer claw member, such that the inner fingers of the first inner claw member can flex towards or away from the outer fingers of the first outer claw member and the inner fingers of the second inner claw member can flex towards or away from the outer fingers of the second outer claw member.

9. A hair fastener as claimed in claim **2**, wherein:

by way of dimensional and/or material characteristics, the inner fingers are more flexible than the inner fingers,

the flexural strengths of the outer fingers and the inner fingers are 40-120 MPa and 43.5-130.5 MPa, respectively, and

the pull forces of the outer fingers and the inner fingers are 1-16 lbs and 0.2-1.5 lbs, respectively.

10. A hair fastener as claimed in claim **9**, wherein:

the first and second outer claw members and the first inner claw members are made of different materials, and/or

the first and second outer claw members are made of a material selected from the group consisting of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), polystyrene (PS), polycarbonate (PC) and high impact polystyrene (HIPS), and the first and second inner claw members are made of a material selected from the group consisting of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers/(SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), and polycarbonate (PC).

11. A hair fastener as claimed in claim **2**, wherein the angle of inclination of a longitudinal axis defined by one of the inner fingers with respect to a longitudinal axis defined by a respective adjacent outer finger ranges from 10° to 45°.

12. A hair fastener comprising:

a first outer claw member and a second outer claw member, wherein the first and second outer claw members are pivotably connected at and movable about a rear end thereof,

a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members,

wherein:

the plurality of outer fingers transitions to respective plurality of prongs such that at a default configuration the prongs of the first claw member and the prongs of the second claw member point at opposite directions and assume an interlocking or alternating formation,

the hair fastener further comprises a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively,

and

a clearance is provided between the inner fingers of the first inner claw member and the outer fingers of the first outer claw member, and a clearance is provided between the inner fingers of the second inner claw member and the outer fingers of the second outer claw member, such that the inner fingers of the first inner claw member can flex towards or away from the outer fingers of the first outer claw member and the inner fingers of the second inner claw member can flex towards or away from the outer fingers of the second outer claw member.

13. A hair fastener as claimed in claim **12**, wherein:

the first inner claw member and the second inner claw member are provided with a plurality of inner fingers, and at the default configuration, convergingly and forwardly extend to the front portion of the first and second outer claw members, the angle of inclination of one of the inner fingers with respect to a respective adjacent outer finger ranges from 10° to 45°.

14. A hair fastener as claimed in claim **12**, wherein:

the first and second outer claw members are provided with a pair of handles at the rear end for operating the hair fasteners,

the plurality of prongs defines a front outwardly facing surface of the front portion of the hair fastener,

the outer fingers and the prongs together define a cavity in or between which the first and second inner claw members reside, and

a distal end of the inner fingers terminates at the front outwardly facing surface of the front portion of the hair fastener.

15. A hair fastener as claimed in claim **12**, wherein the number of the outer fingers from the first outer claw member is n, and the number of the inner fingers from the first inner claw members is $n-1$.

16. A hair fastener as claimed in claim **15**, wherein n is 6.

17. A hair fastener as claimed in claim **12**, wherein the outer fingers from the first outer claw member and the respective inner fingers from the first inner claw member are arranged laterally across the hair fastener in an alternating manner, and the outer fingers from the second outer claw member and the respective inner fingers from the second inner claw member are arranged laterally across the hair fastener in an alternating manner.

18. A hair fastener as claim **17**, wherein at the default configuration, the inner fingers and/or the protrusions of the first inner claw member abut at least some of the inner fingers and/or the protrusions of the second inner claw member.

19. A hair fastener as claimed in claim **12**, wherein:

by way of dimensional and/or material characteristics, the inner fingers are more flexible than the inner fingers,

the flexural strengths of the outer fingers and the inner fingers are 40-120 MPa and 43.5-130.5 MPa, respectively, the pull forces of the outer fingers and the inner fingers are 1-16 lb and 0.2-1.5, respectively, and

the first and second outer claw members are made of a material selected from the group of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), polystyrene (PS), polycarbonate (PC) and high impact polystyrene (HIPS), and the first and second inner claw members are made of a material selected from the group of acrylonitrile butadiene styrene (ABS), K-resin®, styrenic block copolymers (SBC), polyoxymethylene (POM), polyamide (PA), polypropylene (PP), and polycarbonate (PC).

20. A hair fastener comprising:

a first outer claw member and a second outer claw member, wherein the first and second outer claw members are pivotably connected at and movable about a rear end thereof,

a plurality of outer fingers divergingly extending from a rear portion to a front portion of the first and second outer claw members,

wherein:

the plurality of outer fingers transitions to respective plurality of prongs such that in a default configuration the prongs of the first claw member and the prongs of the second claw member point at opposite directions and assume an interlocking or alternating formation,

the hair fastener further comprises a first inner claw member and a second inner claw member fixedly secured to opposite inwardly facing surfaces of the rear portion of the first outer claw member and the second outer claw member, respectively,

at least some of the inner fingers from the first inner claw member are provided with protrusions and at least some of the inner fingers from the second inner claw members are provided with corresponding protrusions and generally point towards each other or at opposite directions and assume a cooperating formation, and

a clearance is provided between the inner fingers of the first inner claw member and the outer fingers of the first outer claw member, and a clearance is provided between the inner fingers of the second inner claw member and the outer fingers of the second outer claw member, such that the inner fingers of the first inner claw member can flex towards or away from the outer fingers of the first outer claw member and the inner fingers of the second inner claw member can flex towards or away from the outer fingers of the second outer claw member in the respective clearance.

* * * * *